

WaveletFlow: Spatially-Localized Frequency Mixing for Flow Matching in Turbulence Modeling

Shuai Meng, Anthony Kuang
UC Berkeley, CS280 Project Proposal

shuaim@berkeley.edu, anthonykuang@berkeley.edu

Abstract

Current generative models for turbulent flows miss fine-scale structures (vortices, shear layers) due to spectral bias. FourierFlow [9] boosts high frequencies using Fourier analysis, but this process is globally defined - it cannot tell where fine structures are. We propose **WaveletFlow**, replacing Fourier processing with wavelet-based processing that is both frequency-aware and spatially localized. This selectively enhances details near vortices without amplifying noise elsewhere. We evaluate on PDEBench compressible Navier-Stokes, testing whether spatial locality improves generative turbulence modeling.

1. What Are We Trying to Do?

Turbulent flows contain structures at many sizes, from large currents to tiny vortices. Predicting them is critical for aircraft design, weather forecasting, and oceanography. Traditional simulations are accurate but extremely slow. Generative AI models predict flows much faster, but produce *blurry* results missing fine-scale details.

We want a generative model that faithfully reconstructs *all* scales of turbulence. Concretely: 4 flow snapshots (density, pressure, v_x , v_y at 128×128) $\rightarrow$ next 4 predicted snapshots (Fig. 1c).

2. Why Is This Problem Interesting?

Wang et al. [9] explained why models blur fine details: during training, small-scale (high-frequency) information is corrupted before large-scale structures, so the model reconstructs easy large patterns first and struggles with fine ones - this is **spectral bias**.

Their FourierFlow uses Fourier transforms to boost high-frequency components ($\sim 20\%$ improvement), but Fourier analysis is **spatially global**: it reports “there is high-frequency energy” but cannot say “there is a vortex in the upper-left corner.” Since vortices appear at *specific locations*, this spatial blindness is a missed opportunity.

3. What Is New and Why Will It Work?

We replace FourierFlow’s Fourier Mixing branch with a **Wavelet Mixing** branch. Wavelets decompose signals into components localized in *both* frequency and space (Fig. 1a). Our WM branch applies multi-level 2D DWT, learnable scale-dependent weights, then inverse DWT:

$$\tilde{\mathbf{d}}_j^{(s)} = (\beta_j^{(s)} + \alpha_j^{(s)} \cdot 2^{j\eta}) \cdot \text{MLP}_j^{(s)}(\mathbf{d}_j^{(s)}) \quad (1)$$

where $\mathbf{d}_j^{(s)}$ are wavelet detail coefficients at scale j , sub-band s , with higher scales receiving larger weights via $2^{j\eta}$ while retaining spatial location.

This is well-motivated: WDNO [3] (ICLR 2025) showed wavelet-domain generation helps with abrupt PDE changes; WNO [8] and U-WNO [4] showed wavelet operators outperform Fourier ones; MWT [2] (NeurIPS 2021) achieved $2\text{--}10\times$ improvements. *No prior work* combines wavelets with flow matching for turbulence (Fig. 1b).

4. Experiments, Feasibility, and Ideal Outcome

Dataset. PDEBench [7] compressible Navier-Stokes (Mach 0.1, 1.0), 128×128 , 4ch. **Baselines.** Vanilla CFM [6]+ViViT [1]; FourierFlow [9]; WaveletFlow (ours). **Metrics.** MSE, nRMSE, Max Error, per-wavenumber energy error. **Experiments.** (E1) Main comparison. (E2) Wavelet family ablation. (E3) Depth $J=1, 2, 3$. (E4) Spatial error maps. (E5) DWT vs. FFT cost. For E1, we report all metrics on both Mach regimes separately, since Mach 1.0 flows contain stronger shocks and finer structures that should amplify any locality advantage. E4 is especially diagnostic: by plotting per-pixel absolute error maps we can visually confirm whether WaveletFlow’s improvements concentrate near vortex cores and shear layers, directly validating the spatial-locality hypothesis. **Timeline (6wk).** Wk1–2: Pipeline+baselines. Wk3–4: WM branch (PyWavelets), train, E1. Wk5: Ablations. Wk6: Report. All on 1–2 A100s. FourierFlow code is public. **Ideal outcome.** WaveletFlow matches/exceeds FourierFlow with particular gains on localized structures.

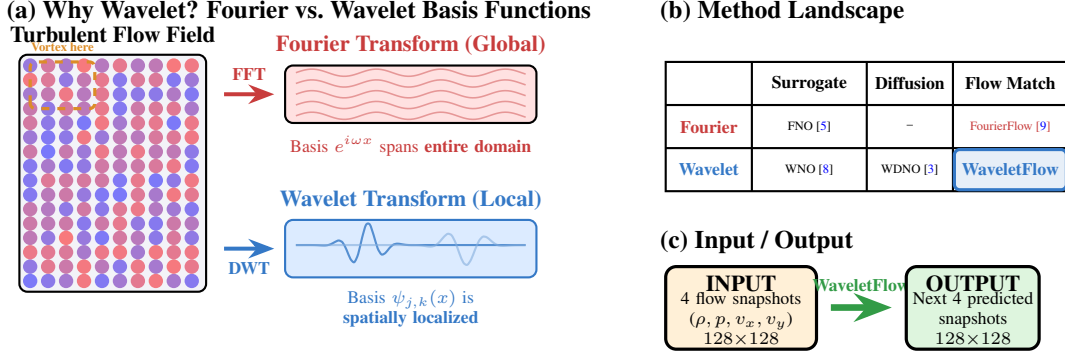


Figure 1. **Motivation and Problem Setting.** (a) Fourier vs. Wavelet transforms. (b) WaveletFlow fills the gap in the method landscape. (c) Input/Output dimensions.

(d) **WaveletFlow Architecture (only the blue Wavelet Mixing branch is new; rest is from FourierFlow)**

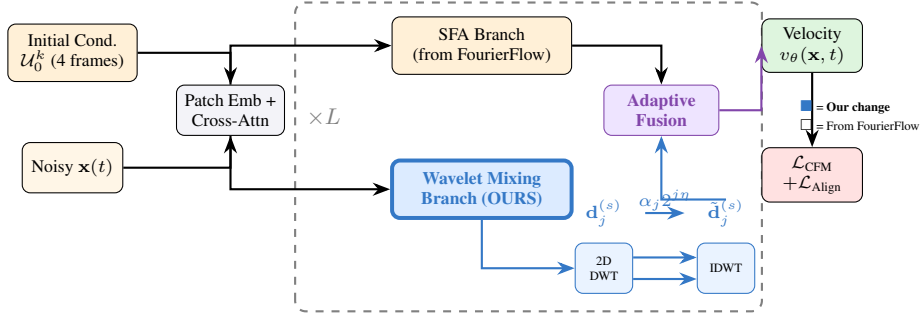


Figure 2. **Architecture details.** We replace the Fourier Mixing branch with our localized Wavelet Mixing branch. The rest of the backbone remains consistent with FourierFlow.

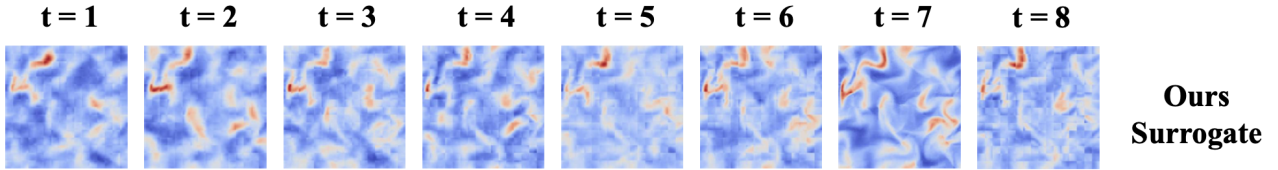


Figure 3. **Temporal evolution of the turbulent flow field.** The sequence shows the progression of the fluid state from $t = 1$ to $t = 8$, highlighting the dynamic vortex structures and localized variations captured by the model.

References

- [1] Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. Vivit: A video vision transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6836–6846, 2021.
- [2] Gaurav Gupta, Xiongye Xiao, and Paul Bogdan. Multiwavelet-based operator learning for differential equations. *Advances in neural information processing systems*, 34:24048–24062, 2021.
- [3] Peiyan Hu, Rui Wang, Xiang Zheng, Tao Zhang, Haodong Feng, Ruiqi Feng, Long Wei, Yue Wang, Zhi-Ming Ma, and Tailin Wu. Wavelet diffusion neural operator. *arXiv preprint arXiv:2412.04833*, 2024.
- [4] Wei-Min Lei and Hou-Biao Li. U-wno: U-net enhanced wavelet neural operator for solving parametric partial differential equations. *Computers & Mathematics with Applications*, 194:272–287, 2025.
- [5] Zongyi Li, Nikola Kovachki, Kamyar Azizzadenesheli, Burigede Liu, Kaushik Bhattacharya, Andrew Stuart, and Anima Anandkumar. Fourier neural operator for parametric partial differential equations. *arXiv preprint*

arXiv:2010.08895, 2020.

- [6] Yaron Lipman, Ricky TQ Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022.
- [7] Makoto Takamoto, Timothy Praditia, Raphael Leiteritz, Daniel MacKinlay, Francesco Alesiani, Dirk Pflüger, and Mathias Niepert. Pdebench: An extensive benchmark for scientific machine learning. *Advances in neural information processing systems*, 35:1596–1611, 2022.
- [8] Tapas Tripura and Souvik Chakraborty. Wavelet neural operator: a neural operator for parametric partial differential equations. *arXiv preprint arXiv:2205.02191*, 2022.
- [9] Haixin Wang, Jiashu Pan, Hao Wu, Fan Zhang, and Tailin Wu. Fourierflow: Frequency-aware flow matching for generative turbulence modeling. *arXiv preprint arXiv:2506.00862*, 2025.